

CHAPTER - 4

FUNCTIONAL DEPENDENCIES AND ANALYSIS OF REDUNDANCY AND DATA ANOMALIES

4.1 INTRODUCTION

Functional dependencies describe the relationship between attributes in a database table, where one attribute uniquely determines another. They play a vital role in designing an efficient database by ensuring data consistency, reducing redundancy, and preventing anomalies. In the AJIO E-Commerce Database Management System, analyzing functional dependencies helps identify how data is organized across different entities and ensures that each table follows normalization principles for efficient storage and reliable data management.

4.2 CUSTOMER TABLE

Attribute	Description
Customer_ID	Unique customer ID
First_Name	Customer's first name
Last_Name	Customer's last name
Email	Customer email address
Phone_Number	Customer contact number
Password	Customer account password
Gender	Customer gender
Date_of_Birth	Customer date of birth
Address	Customer delivery address
City	Customer city
State	Customer state
Pincode	Customer postal code
Registration_Date	Date of account creation

Functional Dependency

Customer_ID → First_Name, Last_Name, Email, Phone_Number, Password, Gender, Date_of_Birth, Address, City, State, Pincode, Registration_Date

The Customer_ID uniquely identifies all customer information.

Redundancy Analysis

Customer information is stored only once using the Customer_ID, which eliminates duplicate customer records and reduces unnecessary data repetition.

Data Anomalies

No significant data anomalies because customer information is stored only once using the Customer_ID.

4.3 PRODUCT TABLE

Attribute	Description
Product_ID	Unique product ID
Product_Name	Product name
Brand	Product brand
Category_ID	Product category ID
Description	Product description
Price	Product price
Discount	Product discount
Stock_Quantity	Available stock
Size	Product size
Color	Product color
Image_URL	Product image
Rating	Customer rating

Functional Dependency

Product_ID → Product_Name, Brand, Category_ID, Description, Price, Discount, Stock_Quantity, Size, Color, Image_URL, Rating

The Product_ID uniquely determines all product details.

Redundancy Analysis

Each product is stored only once, preventing duplicate product information and reducing storage space.

Data Anomalies

No significant data anomalies because each product is uniquely identified by Product_ID.

4.4 CATEGORY TABLE

Attribute	Description
Category_ID	Unique category ID
Category_Name	Category name
Category_Description	Category description

Functional Dependency

Category_ID → Category_Name, Category_Description

The Category_ID uniquely identifies each product category.

Redundancy Analysis

Category details are stored only once and referenced by multiple products through Category_ID.

Data Anomalies

No significant data anomalies because category details are stored separately and referenced using Category_ID..

4.5 ORDER TABLE

Attribute	Description
Order_ID	Unique order ID
Customer_ID	Customer ID
Order_Date	Date of order
Total_Amount	Total order amount
Order_Status	Current order status
Shipping_Address	Delivery address
Delivery_Date	Expected delivery date

Functional Dependency

Order_ID → Customer_ID, Order_Date, Total_Amount, Order_Status, Shipping_Address, Delivery_Date

The Order_ID uniquely identifies every customer order.

Redundancy Analysis

Each order record is stored only once, preventing duplicate order information.

Data Anomalies

No significant data anomalies because each order is uniquely identified by Order_ID.

4.6 ORDER ITEM TABLE

Attribute	Description
Order_Detail_ID	Unique order item ID
Order_ID	Order ID
Product_ID	Product ID

Quantity	Quantity ordered
Unit_Price	Price per product
Total_Price	Total price

Functional Dependency

Order_Detail_ID → Order_ID, Product_ID, Quantity, Unit_Price, Total_Price

The Order_Detail_ID uniquely identifies each item in an order.

Redundancy Analysis

Each ordered product is stored separately, avoiding repeated product entries within an order.

Data Anomalies

No significant data anomalies because each order item is stored separately using Order_Detail_ID.

4.7 PAYMENT TABLE

Attribute	Description
Payment_ID	Unique payment ID
Order_ID	Order ID
Payment_Method	Payment method
Payment_Date	Date of payment
Payment_Status	Payment status
Amount	Amount paid
Transaction_ID	Payment transaction ID

Functional Dependency

Payment_ID → Order_ID, Payment_Method, Payment_Date, Payment_Status, Amount, Transaction_ID

The Payment_ID uniquely identifies each payment transaction.

Redundancy Analysis

Payment details are stored separately for each transaction, eliminating duplicate payment records.

Data Anomalies

No significant data anomalies because payment details are maintained separately using Payment_ID.

4.8 DELIVERY TABLE

Attribute	Description
Delivery_ID	Unique delivery ID
Order_ID	Order ID
Delivery_Partner	Delivery service provider
Tracking_Number	Shipment tracking number
Delivery_Status	Delivery status
Expected_Delivery_Date	Expected delivery date
Actual_Delivery_Date	Actual delivery date

Functional Dependency

Delivery_id → order_id, delivery_partner, tracking_number, delivery_status, expected_delivery_date, actual_delivery_date

The Delivery_ID uniquely determines all shipment details.

Redundancy Analysis

Delivery information is maintained separately, reducing repeated shipment records.

Data Anomalies

No significant data anomalies because delivery information is stored separately using Delivery_ID.

4.9 SELLER TABLE

Attribute	Description
Seller_ID	Unique seller ID
Seller_Name	Seller name
Email	Seller email
Phone_Number	Seller contact number
Address	Seller address
GST_Number	Seller GST number

Functional Dependency

Seller_ID → Seller_Name, Email, Phone_Number, Address, GST_Number

The Seller_ID uniquely identifies each seller.

Redundancy Analysis

Seller information is stored once and linked with multiple products, avoiding duplicate seller records.

Data Anomalies

No significant data anomalies because seller details are maintained in a separate table using Seller_ID

4.10 REVIEW TABLE

Attribute	Description
Review_ID	Unique review ID
Customer_ID	Customer ID
Product_ID	Product ID
Rating	Product rating
Review_Text	Customer review
Review_Date	Date of review

Functional Dependency

Review_ID → Customer_ID, Product_ID, Rating, Review_Text, Review_Date

The Review_ID uniquely identifies every customer review.

Redundancy Analysis

Each review is stored once, preventing duplicate review entries.

Data Anomalies

No significant data anomalies because each review is uniquely identified by Review_ID.

4.11 WISHLIST TABLE

Attribute	Description
Wishlist_ID	Unique wishlist ID
Customer_ID	Customer ID
Product_ID	Product ID
Date_Added	Date product added to wishlist

Functional Dependency

Wishlist_ID → Customer_ID, Product_ID, Date_Added

The Wishlist_ID uniquely identifies each wishlist record.

Redundancy Analysis

Wishlist entries are stored individually, preventing duplicate wishlist records.

Data Anomalies

No significant data anomalies because wishlist entries are stored separately using Wishlist_ID.

4.12 CONCLUSION

The functional dependencies identified for each entity ensure that every non-key attribute is fully dependent on its primary key, resulting in a well-structured and normalized database. This design minimizes data redundancy, maintains data integrity, and prevents insertion, update, and deletion anomalies. As a result, the AJIO E-Commerce Database Management System provides efficient data storage, reliable transaction processing, and improved overall system performance, making it suitable for handling large-scale online shopping operations.